EQ2341 Project Report: Activity Recognition using Hidden Markov Models (HMM)

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1 Introduction

The primary objective of this project is to implement an accurate sequence classifier for human activity recognition based on data acquired from a smartphone's internal accelerometer. Activity recognition is very useful since it is a foundational technology for context-aware computing, fitness tracking, and mobile health monitoring.

The algorithm is designed to accurately discriminate between three physical activities; standing still, walking, and running. The main challenge was identifying these states within a single, continuous data stream where the user dynamically shifts between activities. The Hidden Markov Model (HMM) framework is very suitable for this task since it models both the instantaneous sensor readings and the temporal dependencies of human motion. This reduces sporadic classification errors compared to analyzing the data one frame at a time.

2 Methodology

2.1 Data Recording and Pre-processing

Data was collected using a mobile application that records raw tri-axial accelerometer data. The sensor output is represented as a time series of three-dimensional vectors $x_t \in \mathbb{R}^3$, where each component represents acceleration along the x, y and z axes in Euclidean space.

To create a robust dataset, multiple isolated instances of each activity were recorded (standing still, walking and running). A separate, continuous recording (test data) containing dynamic transitions between all three activities was captured for evaluation.

Raw accelerometer data is susceptible to high-frequency sensor noise. Therefore, the Simple Moving Average (SMA) filter with a window size of $W = 2$ frames was applied. This specific window size was chosen to smooth out microscopic jitters/gaps while preserving the sharp transition boundaries and the distinct variance amplitudes that characterize running and walking.

2.2 Hidden Markov Model Architecture

The system is modeled as a discrete-time Markov chain with $N = 3$ hidden states. Because sensor observations are continuous variables, the emission probabilities $P(x_t|S_t = j)$ was modeled using parametric multivariate Gaussian distributions with full covariance capabilities.

The HMM topology was initialized with uniform starting probabilities ($\pi_i = 1/3$) and a strongly diagonal transition matrix ($a_{ii} = 0.98$). This initial transition matrix forced the model to stick to one activity and not rapidly switch between activities during early training.

2.3 Parameter Estimation (Baum-Welch EM Algorithm)

Rather than manually calculating the parameters for each state, the Expectation-Maximization (Baum-Welch) algorithm was utilized. To help the algorithm start learning, it was given three separate starting guesses based on how gravity pulls on the phone. The EM algorithm iteratively refined the model parameters (π, A, μ, Σ) over 10 iterations to maximize the log-likelihood of the concatenated training sequences.

2.4 Sequence Decoding (Viterbi Algorithm)

Following training, the Viterbi algorithm was implemented to decode the most probable sequence of hidden activities for the unseen test data.

3 Experimental Results

This section presents the optimized model parameters generated by the EM algorithm and evaluates the classification performance on the continuous test sequence.

3.1 EM Optimization Results

After 10 iterations, the EM algorithm converged successfully. The model autonomously learned to map the three hidden states to our physical activities by analyzing the variance in the accelerometer data. It perfectly mapped the standing state, but struggled to clearly separate the dynamic states (walking and running).

The optimized state transition matrix (A) retained strong diagonal probabilities, indicating stable state duration:

$$A = \begin{bmatrix} 0.969 & 0.005 & 0.027 \\ 0.000 & 0.960 & 0.040 \\ 0.063 & 0.091 & 0.846 \end{bmatrix} \tag{1}$$

The optimized Gaussian parameters (Means and Covariance Diagonals) clearly distinguish the three states, as detailed in Table 1.

Table 1: Optimized Gaussian Parameters (Means and Covariance Diagonals) after EM training.

Activity State	μ_x	μ_y	μ_z	σ_x^2	σ_y^2	σ_z^2
Standing (State 0)	0.213	-0.950	0.096	0.006	0.007	0.014
Walking (State 1)	-0.169	-0.705	-0.322	0.064	0.068	0.067
Running (State 2)	-0.060	-1.231	0.054	0.102	0.726	0.151

3.2 Classification on Unseen Test Data

The trained model was evaluated on a continuous test sequence consisting of 483 samples. The total log-probability of the test sequence fit was -179.18, which indicates a strong overall statistical fit.

The Viterbi decoding successfully and perfectly isolated the "Standing Still" activity, demonstrating a solid, uninterrupted block of correct classification between 42s and 66s. However, the timeline reveals that the model struggled to clearly distinguish between "Walking" and "Running" during the initial dynamic phase (0s to 40s), where the predictions flickered back and forth between the two states. The model eventually stabilized during the final 28 seconds (68s to 96s), locking into a continuous "Walking" classification.

4 Analysis and Discussion

4.1 Variance as the Primary Discriminator

The results in Table 1 reveal how the EM algorithm clustered the activities. Since a smartphone experiences approximately $1G$ of downward acceleration regardless of the activity, the mean vectors (μ) for all three states remained relatively close to each other in Euclidean space. Instead, the model relied heavily on the covariance diagonals (σ^2) to differentiate the states.

State 0 exhibits near-zero variance (≈ 0.007), representing the rigid stability of standing still. State 1 shows moderate variance (≈ 0.06) consistent with the rhythmic impact of walking. Finally, State 2 exhibits extreme variance, particularly on the y-axis ($\sigma_y^2 = 0.726$), capturing the high-impact vertical shocks associated with running.

4.2 Transition Stability and Viterbi Decoding

A common issue in HMM-based sensor classification is rapid flickering between activities. For example, the exact moment a foot hits the ground might temporarily look like standing still. Our Viterbi output strongly resisted this flickering. This stability comes directly from the optimized transition matrix. Because the self-transition probabilities for standing and walking remained above 0.95, the model had a strong mathematical preference to stay in its current state. This allowed it to ignore brief, noisy sensor readings instead of rapidly switching activities.

4.3 Classifier Confusion: Walking vs. Running

As seen in the Viterbi output, the model frequently confused walking and running during the first 40 seconds of the test sequence. This confusion is physically intuitive. Both are cyclic, dynamic activities. If a user performs a brisk walk with heavy steps, or a slow jog, the variance profiles of the accelerometer data overlap significantly. Because this specific HMM setup relies strictly on spatial distribution (variance amplitudes) rather than frequency-domain features (such as step frequency), the model lacks the necessary contextual data to confidently separate high-energy walking from low-energy running.

4.4 Limitations and Future Work

While the model performs well, the variance-based classification is inherently sensitive to device placement. In this study, the phone's orientation was kept relatively stable. If the user were to change the device's orientation (e.g., moving it from a hand to a loose pocket), the axis-specific Gaussian models might fail to correctly calculate the likelihoods. Future iterations of this project could improve robustness by pre-processing the 3D vector into an orientation-invariant feature, such as the total acceleration magnitude ($\sqrt{x^2 + y^2 + z^2}$).

5 Conclusions

This project successfully implemented a Hidden Markov Model classifier capable of continuous human activity recognition. By initializing strategic anchors and leveraging the Baum-Welch EM algorithm, the model managed to cluster tri-axial accelerometer data into standing, walking and running states based predominantly on distinct variance profiles. While the subsequent Viterbi decoding demonstrated robust, flicker-free classification for the static state, the results revealed an expected ambiguity between walking and running due to their overlapping physical variance.